

***B.Tech. Degree VIII Semester (Supplementary)
Examination, September 2008***

ME 802 MACHINE DESIGN II

(2002 Scheme)

Time: 3 Hours*

Maximum Marks: 100

(Design data book permitted. Assume reasonable data wherever necessary)

- I
- a) Describe any two internal expansion elements in automotive shoe brakes. (8)
- b) Design a belt to transmit 50 HP at 740 rpm to a rolling machine. Speed reduction desired is 3:1, distance between pulleys is 3.5 meters and diameter of the rolling machine pulley is 1.5m. It is desired to run fabric belt because of the conditions in the rolling mill and cast iron pulley. (17)

OR

- II
- a) With neat sketch explain the working of plate clutch. (8)
- b) A multi disc clutch employs 3 steel and 2 bronze disc having outer diameter 30 cm and inner diameter 20 cm. If the coefficient of friction is assumed as 0.22, find
- i) the axial pressure and HP transmitted if the unit normal pressure is 1.2 kg/cm^2 and the rpm is 750.
- ii) the axial pressure of the unit normal pressure in kg/cm^2 if the clutch transmit 30 hp at 500 rpm. (17)

- III
- a) What are the commonly used gear materials.? (4)
- b) Explain the following terms used in Bevel gear : (4)
- i) Cone distance ii) Pitch angle
- iii) Face angle iv) Root angle
- c) Design a set of helical gear to transmit 80hp to effect a speed reduction of 2.5:1 with pinion connected to motor rotating at 1440 rpm. The teeth of the gears are of full depth 20° pressure angle in the normal plane. Peripheral pitch line velocity may be limited to 600m/mts. Assume gear and pinion are to be made of forged steel having ultimate tensile strength of 6000 kg/cm². The pinion properly heat treated to obtain hardness of 240 BHN. Assume service factor of 2 and number of teeth in pinion = 16. (17)

OR

- IV
- a) What is interference in gears? (4)
- b) What do you understand by the term :
- i) Circular pitch ii) Diametrical pitch
- iii) Module iv) Backlash (4)
- c) Two shafts at right angles to each other one connected by a pair of $14\frac{1}{2}^\circ$ involute teeth bevel gears. The velocity ratio is 3:1. The pinion is made of steel having an allowable static stress = 900 kg/cm^2 and the gear is of cast steel having the allowable static stress = 700 kg/cm^2 . The pinion transmits 45 hp at 1200 rpm. If the pinion has 30 teeth determine the mean module, face width and pitch diameter of the gear wheel from the stand point of strength taking velocity factor into consideration. Assume face width to be $\frac{1}{4}$ th of the slant height of the pitch cone

The tooth form factor = $\Psi = \pi \left(0.124 - \frac{0.686}{mt} \right)$, where mt is the formative

number of teeth. The velocity factor = $\frac{280}{280 + V}$ where V is the pitch line velocity in m/mt.

If the pinion shaft over hangs by 12.5 cm, determine its diameter for the safe shear stress of 400 kg/cm².

(17)
(Turn Over)

- V
- a) What are the various types of lubrication method for journal bearings? (4)
 - b) Explain hydro dynamic theory of lubrication. (4)
 - c) Calculate the dynamic load rating of a deep groove ball bearing which carries a radial load of 360 kg and a thrust load of 300 kg at 1750 rpm. The service is 8 hr/day and the bearing is to be designed for life of 18000 hours with no more than 10% failures. (17)
- OR**
- VI
- a) Compare roller bearing with journal bearing. (4)
 - b) What is meant by static and dynamic capacity of bearing? (4)
 - c) For a bearing the equivalent load is 1000 kg at a speed of 1000 rpm for 3800 hours average life. Calculate the corresponding basic dynamic load rating for 1 million revolution. (17)
- VII
- a) Explain the general design recommendations for the following:
 - i) Forging
 - ii) Milling
 - iii) Casting
 (15)
 - b) Explain tolerance. (5)
 - c) How surface roughness is measured? (5)
- OR**
- VIII
- a) Explain general design recommendation for i) Weld joint ii) Rolled sections (10)
 - b) Explain factors to be considered by an engineer while designing a casting. (10)
 - c) Explain Limit and Fits. (5)

